Perfect t-embeddings of Uniform Hexagons

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(joint work with T. Berggren and M. Russkikh)

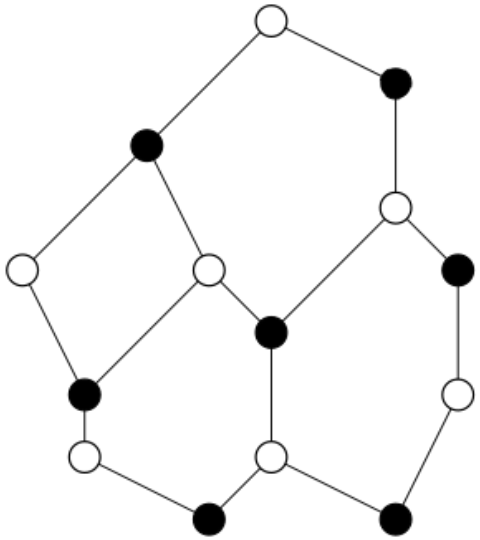
Outline

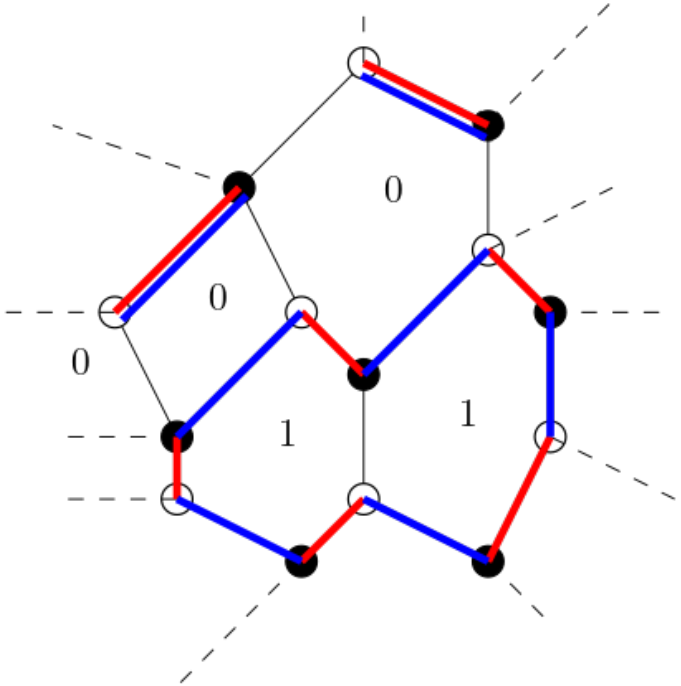
1. Dimer model background and t-embedding motivation
2. t-embeddings in general
3. Perfect t-embeddings of Hexagons, GFF for lozenge tilings

Dimer model basics

Dimer model

1. We study random perfect matchings M of a bipartite planar weighted graph $(\mathcal{G}, \nu)$. The probability measure is $\mu_{\mathcal{G}}(M) = \frac{1}{Z} \prod_{e \in M} \nu_e$.
2. Define the height function $h = h_M$ by superimposing M with a reference matching M_0 and viewing the loops as level lines of h.
3. **Goal:** Study asymptotic height fluctuations $h - \mathbb{E}[h]$.

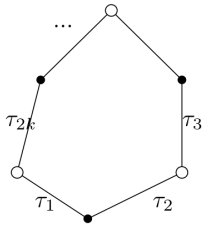




Kasteleyn signs and local statistics

1. **Complex Kasteleyn weighting:** We choose $|\tau_e| = 1, e \in E$ such that around any face

$$\frac{\tau_1}{\tau_2}\frac{\tau_3}{\tau_4}\cdots\frac{\tau_{2k-1}}{\tau_{2k}} = (-1)^{k+1}.$$

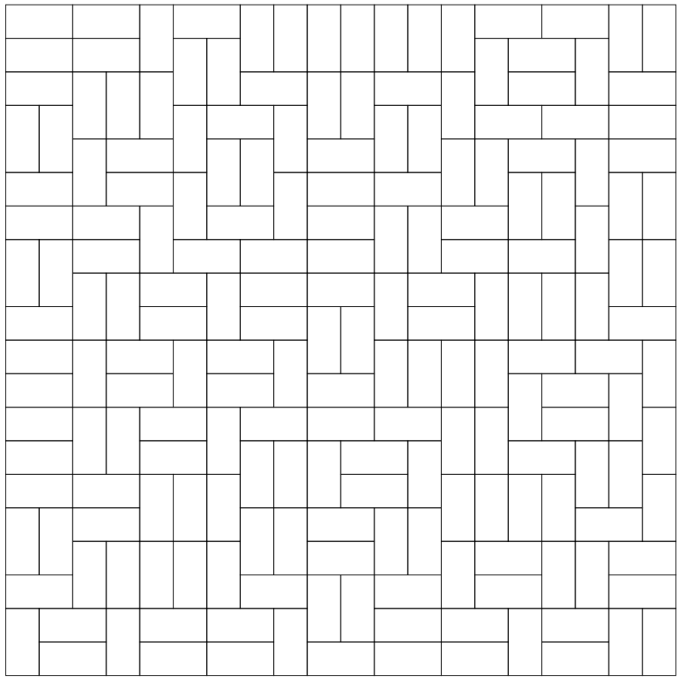


2. Using edge weights $\{\nu_e\}$ and signs $\{\tau_e\}$, a **Percus-Kasteleyn matrix** K (whose rows index white vertices and columns index black vertices) is defined by $K(w, b) = \tau_{bw}\nu_{bw}$.
   - **Theorem [Percus '69, Kasteleyn '61]:** $Z = |\det K|$.
   - **Corollary [Kenyon]:** Let $E' = \{e_i = (b_i, w_i)\}_{i=1}^k$ be a set of edges. Then

$$\mu_{\mathcal{G}}(\{M : E' \subset M\}) = \left(\prod_{i=1}^{k} K(w_i, b_i)\right) \det(K^{-1}(b_i, w_j))_{i,j=1}^k.$$

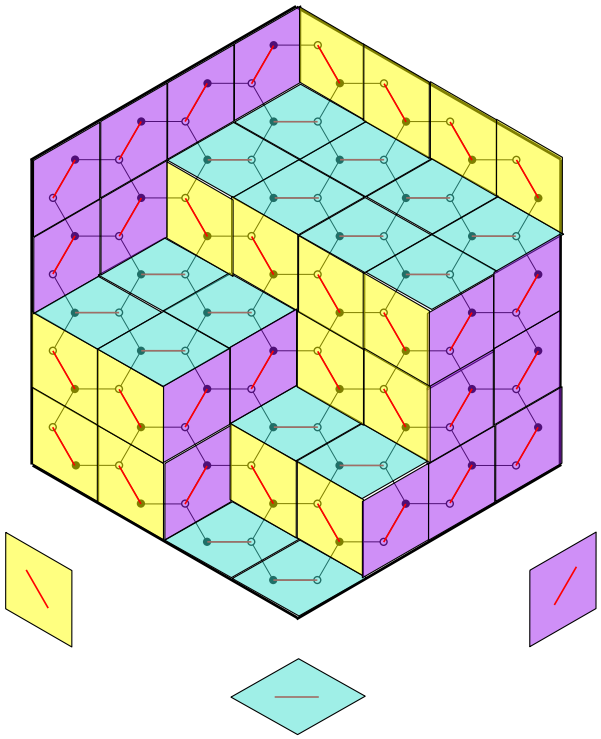
Dimer model on square lattice

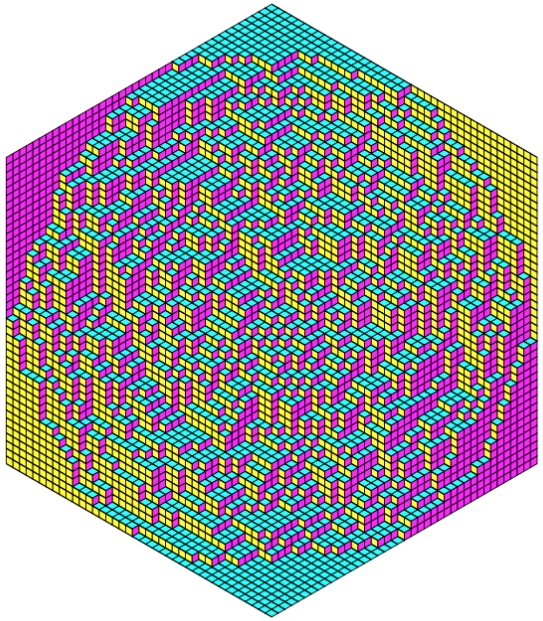
1. Useful point of view ([Kenyon]): K approximates $\overline{\partial}$.
2. In certain setups, this leads to proofs of convergence of height function fluctuations to a *Gaussian free field* [Kenyon'00, Russkikh'18], a *conformally invariant limit*.



General lattices and boundary conditions

1. Useful point of view for "flat" boundary conditions in square lattice: K approximates $\overline{\partial}$.
2. In general, can still approximate $K \sim \overline{\partial}$, though this approximation happens in a nontrivial way. E.g. even at leading order we see *phase separation*.
3. Big goal: use *t-embeddings* to do this.





Fluctuations: a general conjecture

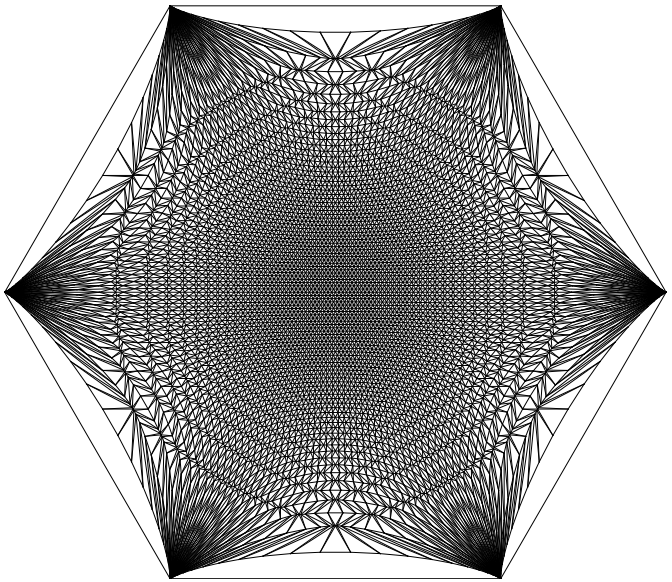
1. **A general conjecture of [Kenyon-Okounkov '07] (dimer models with $\mathbb{Z}^2$-periodic weights):** Height fluctuations converge to a GFF, generically in a nontrivial **complex structure** which depends in a nontrivial way on boundary conditions.
2. **Goal of studying new "t-embeddings"** introduced and studied by [Affolter '18], [Kenyon-Lam-Ramassamy-Russkikh '18]: To develop a general framework to prove convergence to GFF using *discrete holomorphicity*; a theory of discrete holomorphic functions on t-embeddings is developed in [Chelkak-Laslier-Russkikh '20].
3. Theorem of [Chelkak-Laslier-Russkikh '21]: *Perfect t-embeddings* can be used to prove convergence to the GFF in the complex structure of an associated *maximal surface in the Minkowski space*.

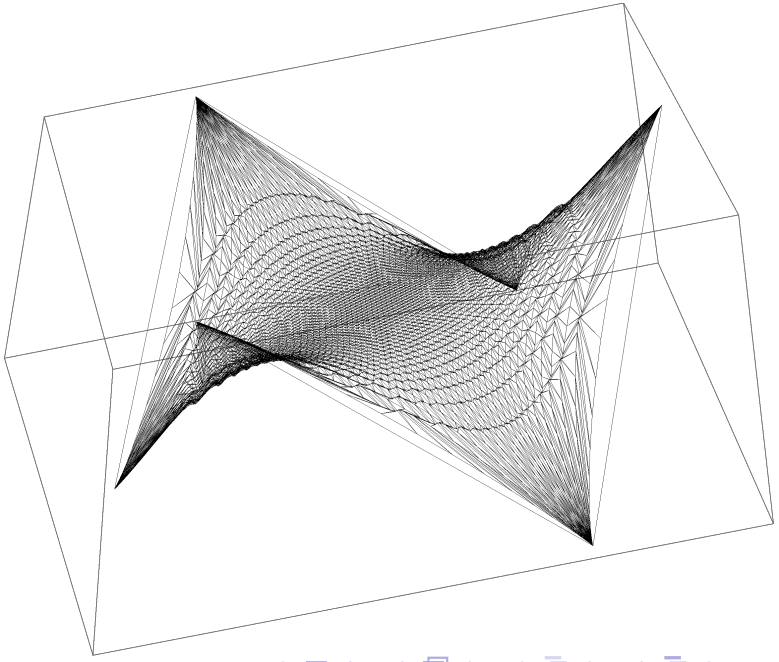
Major questions: existence, and technical assumptions

To use the machinery of [CLR '21] in a given setup, one must

1. **Construct a "perfect" t-embedding** (existence is conjectured).
2. show convergence to a **maximal surface**.
3. show that the **nondegeneracy assumptions** of [CLR'21] hold.

In our work, we complete 1-3 for the uniform hexagon, so we complete a new proof of GFF fluctuations using tools of [CLR '21].





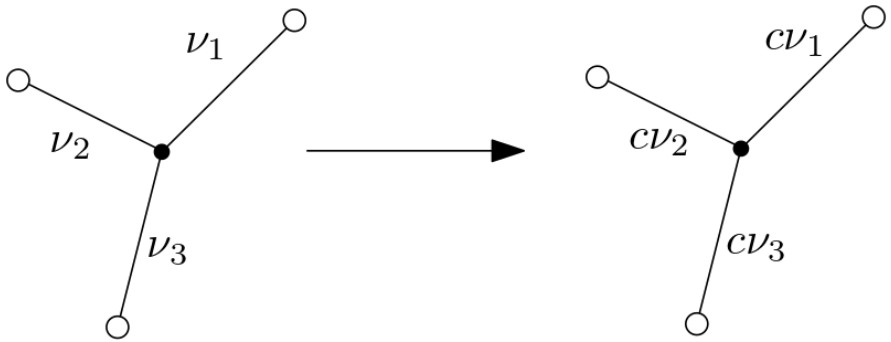
t-embeddings

Gauge equivalence

1. Recall we have a weight function $\nu : E \to \mathbb{R}_{>0}$. The **Boltzmann measure** on dimer configurations is

$$\mu(M) = \frac{1}{Z} \prod_{e \in M} \nu_e.$$

2. Two weight functions are *gauge equivalent* if one can be obtained from the other by a sequence of measure-preserving local operations, as below:

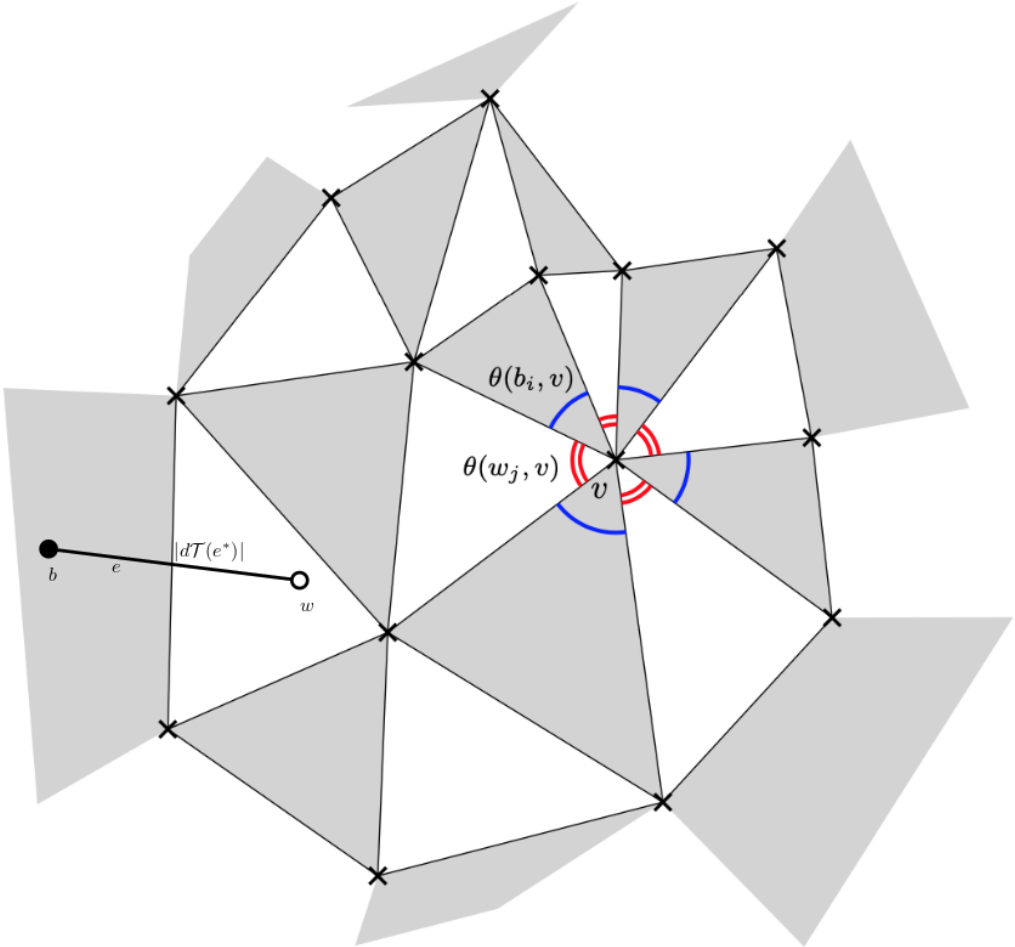


Definition: t-embedding

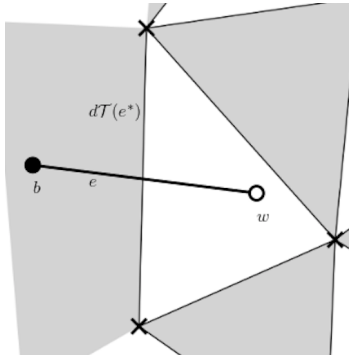
A t-embedding $\mathcal{T} : \mathcal{G}^* \to \mathbb{C}$ of a bipartite graph $(\mathcal{G}, \nu)$ satisfies:

1. **Properness**, **Gauge equivalence** (the edge weights are geometric, and induce the same statistics), **Angle condition**

Figure from M. R.



Geometric Kasteleyn weights



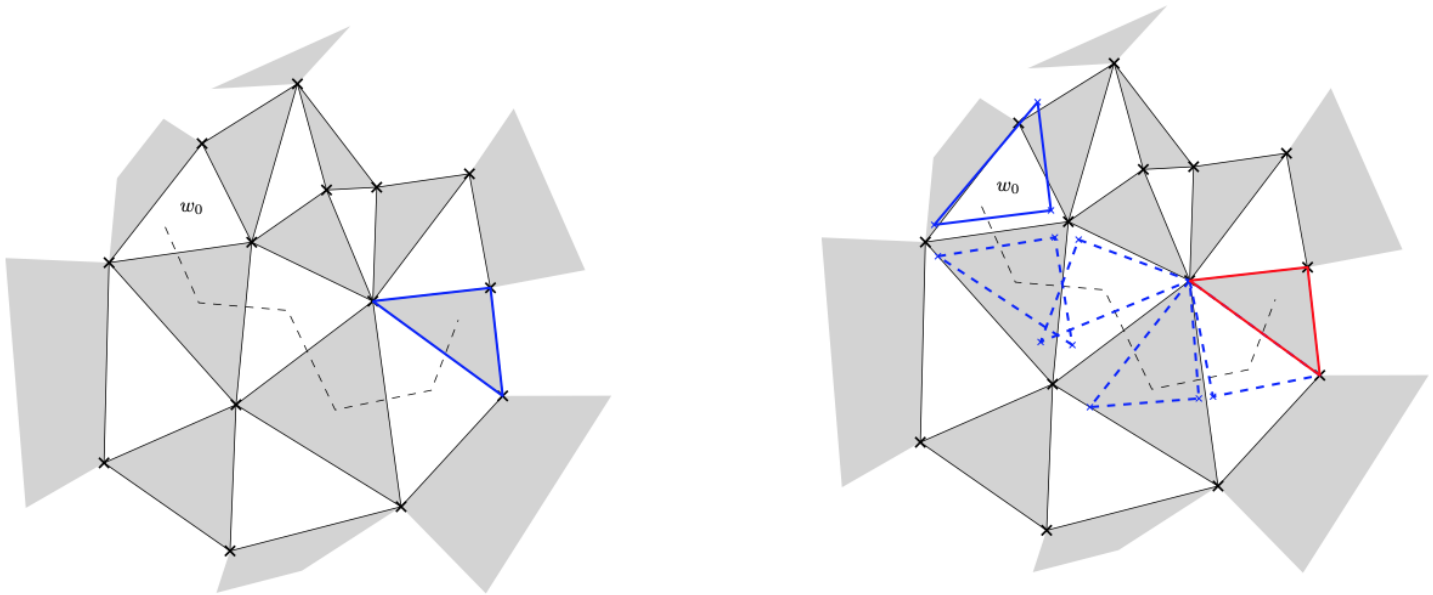
1. A t-embedding defines a Kasteleyn matrix for $\mathcal{G}$: $K_{\mathcal{T}}(w,b) = d\mathcal{T}((wb)^*)$.
2. In fact, if K is a real Kasteleyn matrix of $\mathcal{G}$, the **angle condition** implies existence of *Coulomb gauges* $F(b)$ and $G(w)$ such that

$$d\mathcal{T}((wb)^*) = G(w)K(w,b)F(b)$$

   where $\sum_w G(w)K(w,b) = 0$ (and similarly $KF = 0$).
3. Conversely, may start with F, G in the kernel of K (at interior vertices). The only property not guaranteed is *properness* of $\mathcal{T}$.

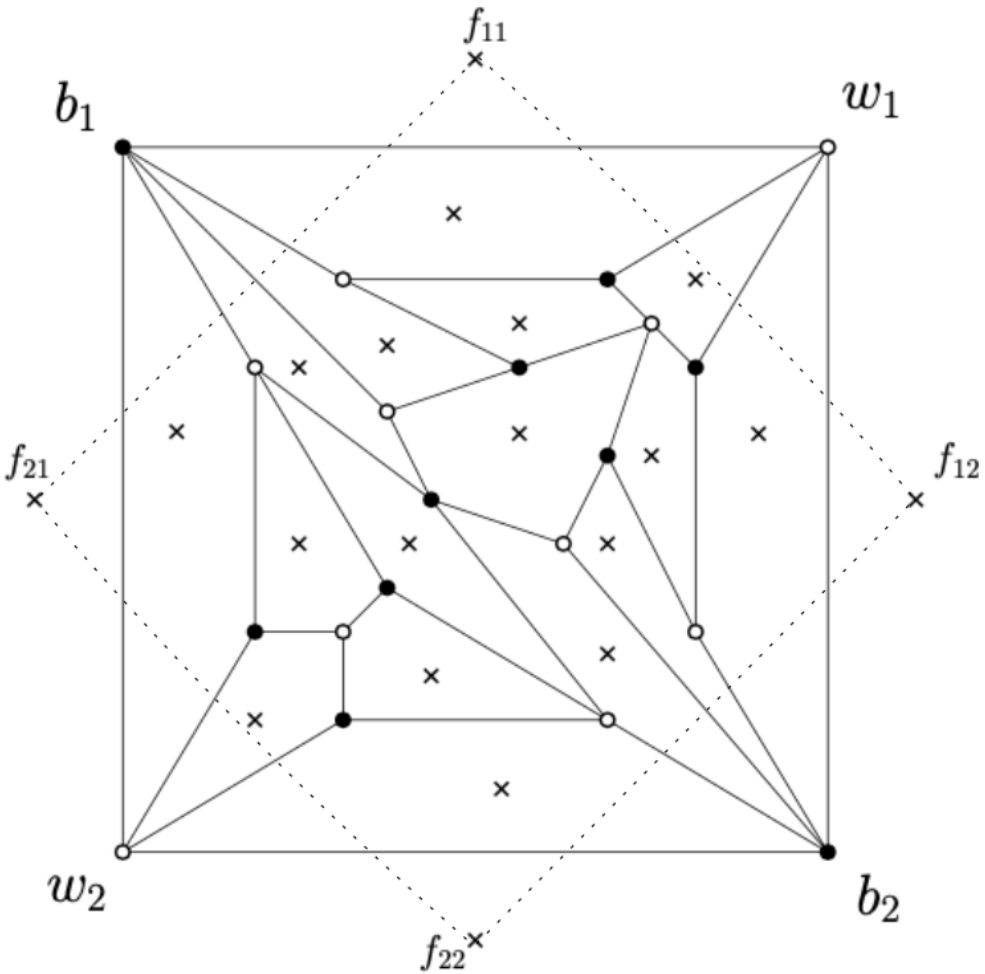
Definition: Origami map

1. The origami map is a piecewise smooth map $\mathbb{C} \supset \mathit{Image}(\mathcal{T}) \to \mathbb{C}$.
2. Geometrically, to compute $\mathcal{O}(b)$ for a face b, fold across edges of $\mathcal{T}$ along a face path to a reference face w_0:



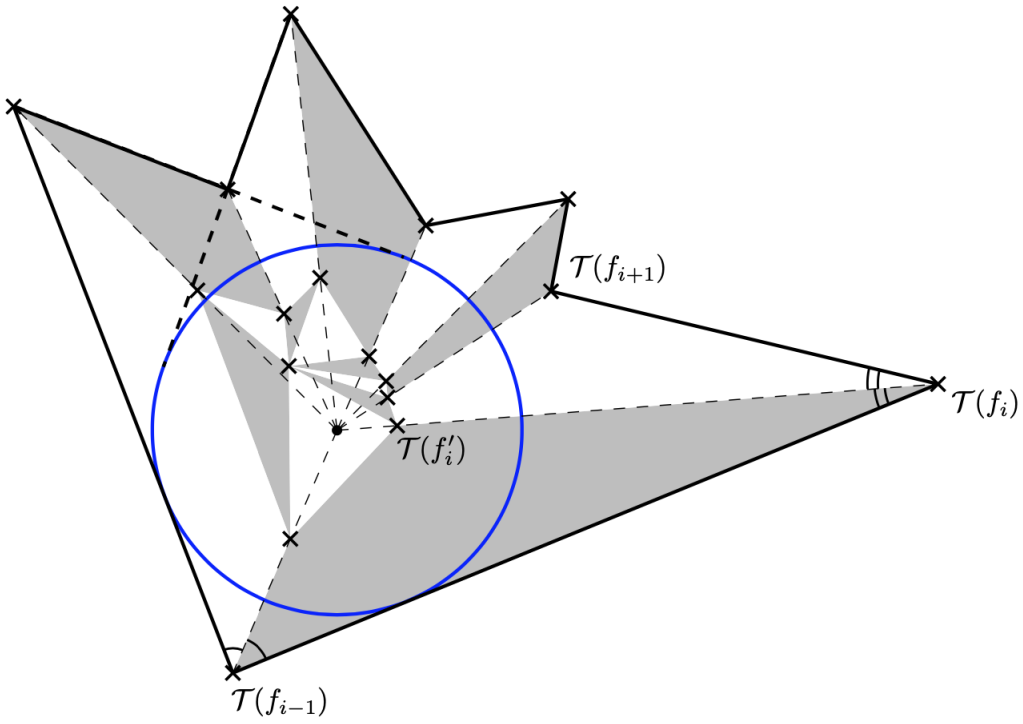
Boundary conditions: Augmented dual graph

1. In fact, we embed the *augmented dual graph*.



Definition: Perfect t-embeddings

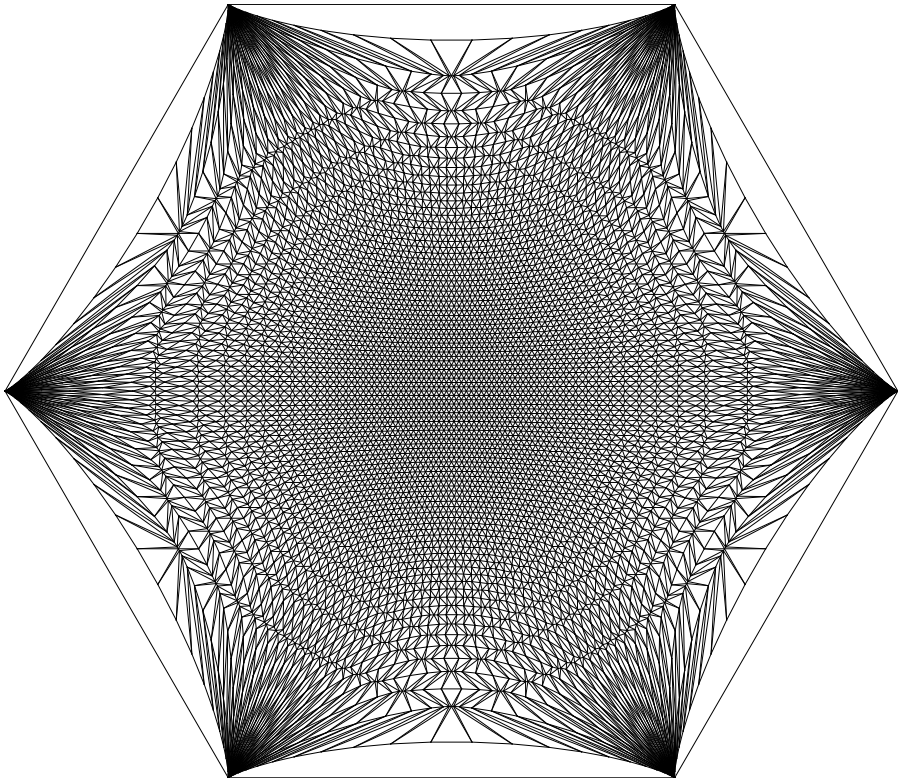
1. Recall we have a boundary polygon P. A t-embedding is **perfect** if
   - P is tangential to the unit disk;
   - The dual edges connecting outer dual vertices to dual vertices in the interior of $\mathcal{G}^*$ are angle bisectors.



Fluctuations: a general theorem [Chelkak, Laslier, Russkikh]

1. Given *perfect* ([CLR '21]) t-embeddings $\mathcal{T}_n$ which satisfy two nondegeneracy assumptions, and that the discrete surfaces $(\mathcal{T}_n, \mathcal{O}_n)$ converge to a **maximal surface in Minkowski space** $S \subset \mathbb{R}^{2,1}$, then height function fluctuations converge to Gaussian Free Field in the conformal structure of S.
2. General existence of perfect t-embeddings is a conjecture. It is known if the outer face has degree 4.
3. Goal: Understand how to construct and analyze perfect t-embeddings.

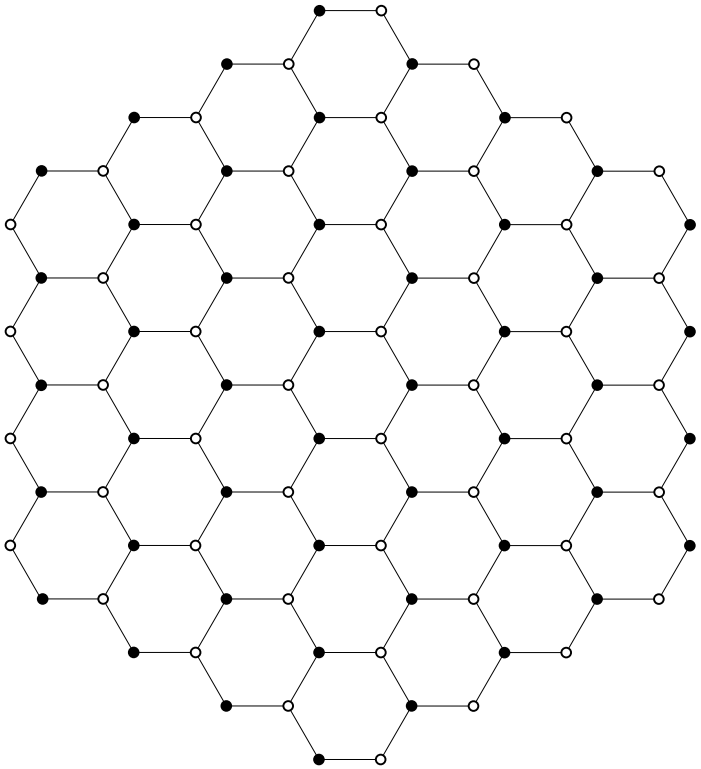
Uniform Hexagon

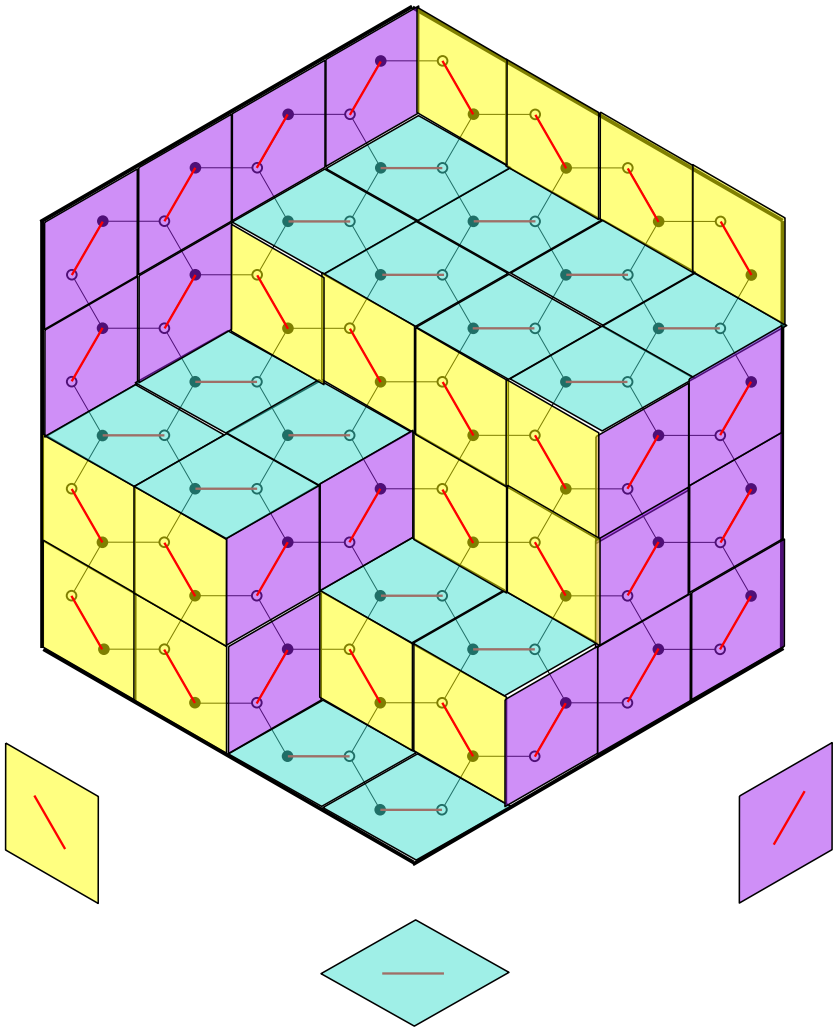


How to build this t-embedding?

Hexagon

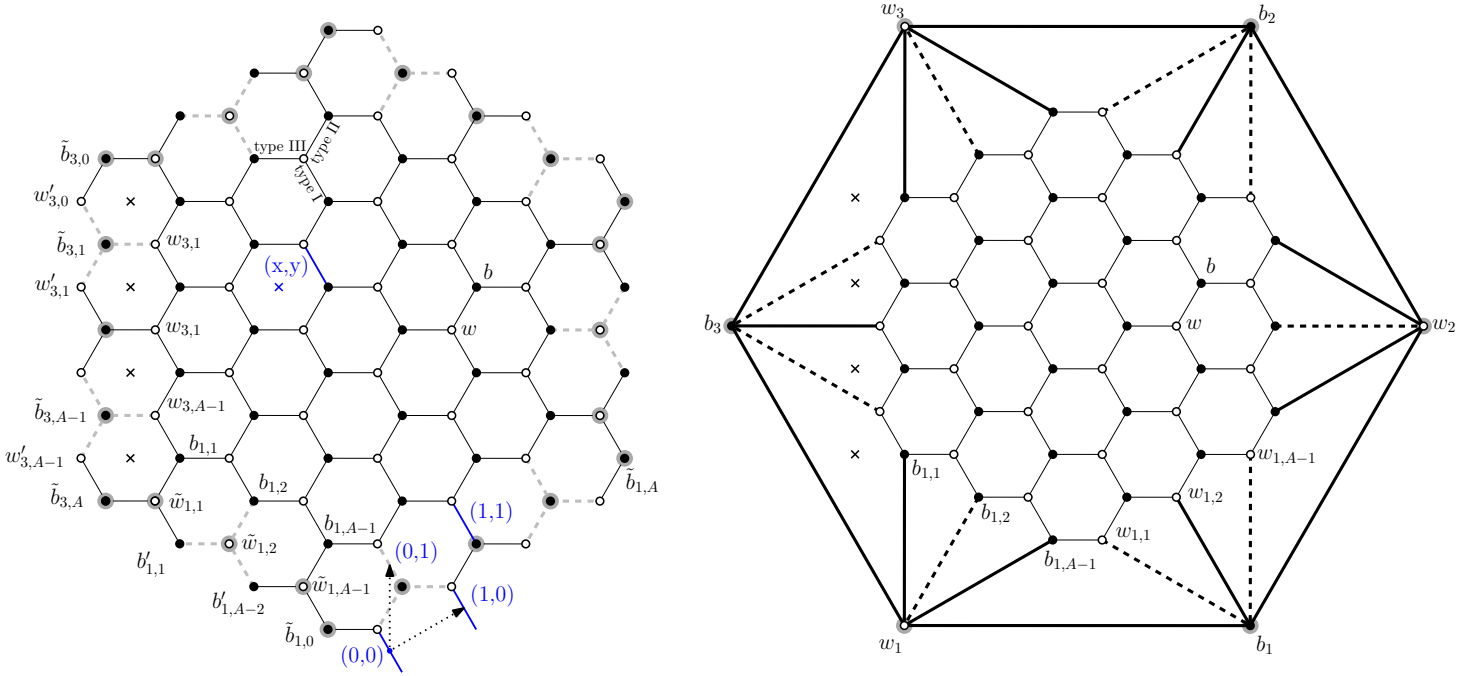
1. We consider the $N \times N \times N$ **Hexagon**: (See below for $N = 4$).
2. Perfect matchings are *lozenge tilings of a hexagon*.





Construction of t-embedding for (reduced) Hexagon

1. We construct an embedding of H'_A, the $A \times A \times A$ *reduced hexagon*.



2. The dimer statistics inside are the same as for uniformly random lozenge tilings.

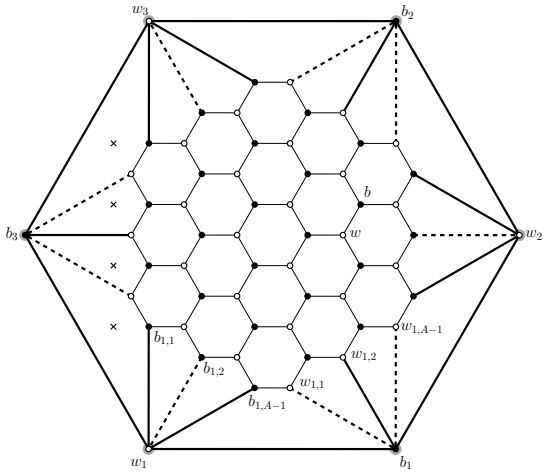
Gauge functions in terms of $K_{\mathbb{R}}^{-1}$

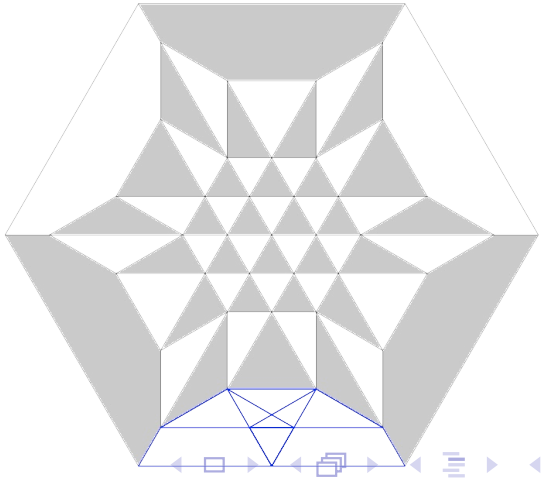
Theorem (Berggren, N., Russkikh)

The Coulomb Gauge functions define a perfect t-embedding of H'_A:

$$G(w) = \sum_{j=1}^{3} e^{-\mathrm{i}j2\pi/3} K_{\mathbb{R}}^{-1}(b_j, w)$$

$$F(b) = -\sum_{j=1}^{3} e^{-\mathrm{i}(j-1)2\pi/3} K_{\mathbb{R}}^{-1}(b_j, w)$$





t-embeddings: a contour integral formula

Using exact formulas of [Petrov '12] for $K_{\mathbb{R}}^{-1}$, we obtain

Theorem (Berggren, N., Russkikh)

For all faces (x, n) in the $A \times A \times A$ hexagon

$$\mathcal{T}_A(x, n) \quad = \quad \frac{1}{(2\pi \mathrm{i})^2} \int_{\mathcal{E}_2} \int_{\mathcal{E}_1} e^{A(S(w)-S(z))} f(w) g(z) \frac{dz\, dw}{z - w}. \quad (1)$$

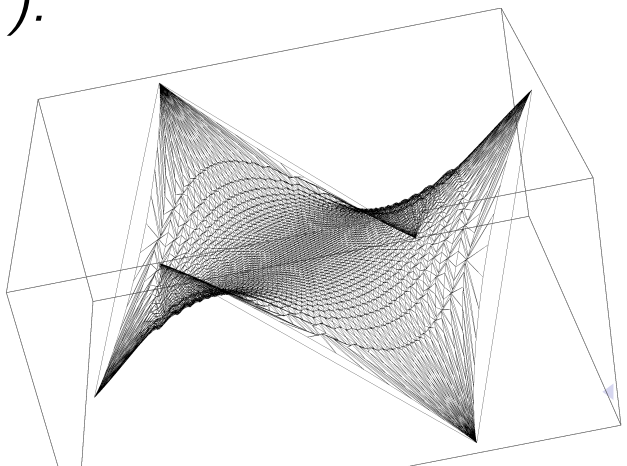
Above $f(w), g(z)$ are rational functions, and $S(z) = S(z; \frac{x}{A}, \frac{n}{A})$, where $S(z; \chi, \eta)$ is the same action function appearing in [Petrov '12]. A similar formula for $\mathcal{O}$ is obtained.

1. This expression is amenable to asymptotic analysis.
2. As shown in [Petrov '12], the action function is related to the conformal structure of the limiting Gaussian free field. The *critical point* $\zeta(\chi, \eta)$ is the uniformizing map.

Asymptotic analysis of t-embeddings: maximal surface

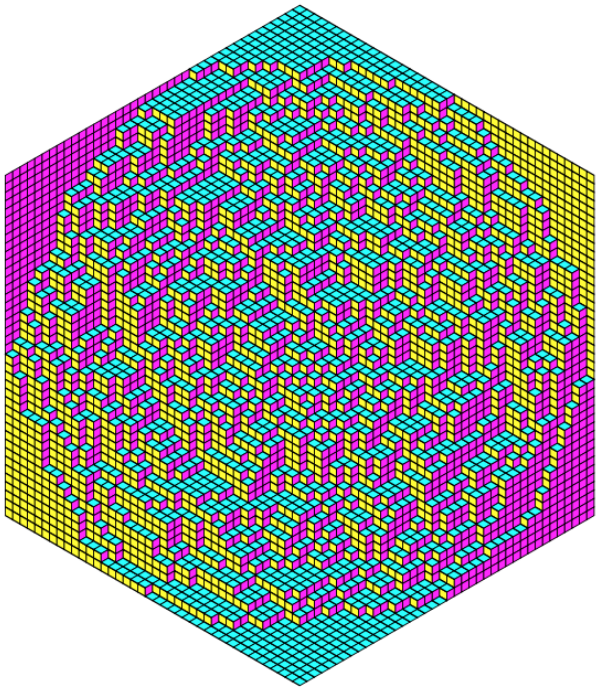
Theorem (Berggren, N., Russkikh)

1. *We have*
$$\lim \mathcal{T}_A(A\chi, A\eta) = z(\chi,\eta) = \frac{1}{2\pi \mathrm{i}} \int_{\overline{\zeta}}^{\zeta} f(z)g(z)dz$$
$$\lim \mathcal{O}_A(A\chi, A\eta) = \vartheta(\chi,\eta) = \frac{1}{2\pi \mathrm{i}} \int_{\overline{\zeta}}^{\zeta} f(z)\overline{g(\overline{z})}dz.$$
where f,g are explicit and $\zeta = \zeta(\chi,\eta)$ solves $\partial_z S(z;\chi,\eta) = 0$.
2. *The limiting graph $\{z, \vartheta(z)\}$ is a maximal surface in $\mathbb{R}^{2,1}$ (has mean-curvature zero with respect to the ambient metric $|dz|^2 - d\vartheta^2$).*



Fluctuations of hexagon height function

1. The *height function* of a matchings is obtained from viewing a tiling as "stacks of cubes in a room".
2. What is known already about lozenge tilings of a hexagon?
3. Limit Shape Theorem [Cohn-Larsen-Propp '97]: $\frac{1}{N} h_N(Nx, Ny)$ converges in probability to a deterministic limit.
4. Fluctuation Theorem [Petrov '12]: $h_N(Nx, Ny) - \mathbb{E} h_N(Nx, Ny) \to GFF$.

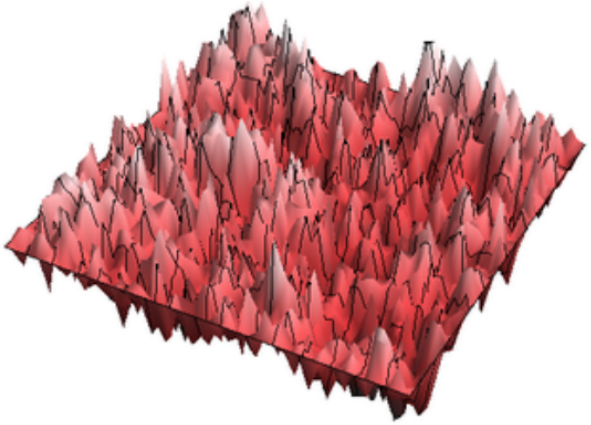


Fluctuations: Gaussian free field

1. The Gaussian free field on a domain $D \subset \mathbb{R}^2$ is the Gaussian process on D with covariance structure given by the Greens function: for any $x_1, \ldots, x_k \in D$, $(F_D(x_i))_{i=1}^k$ is a mean 0 Gaussian vector and

$$\mathbb{E}[F_D(x_i)F_D(x_j)] = G_D(x_i, x_j)$$

   where G_D is the Dirichlet Greens function on D.
2. Note G_D is singular on the diagonal; F_D is really a random distribution (not a function).

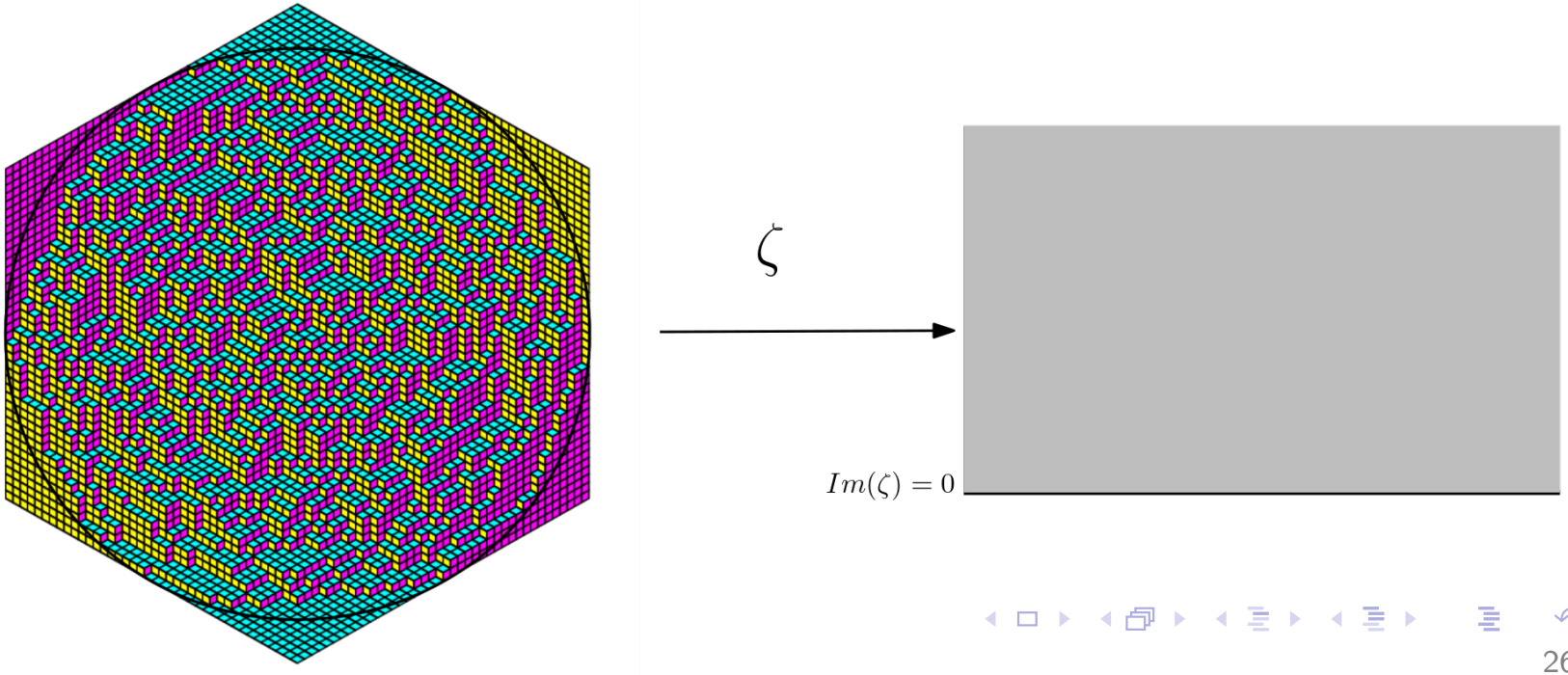


Fluctuations of hexagon height function

1. Exists a diffeomorphism $\zeta : \{(x, y) \in \text{Inscribed disc}\} \to \mathbb{H} \subset \mathbb{C}$, the *uniformizing map*. It is **not a conformal map**.
2. **Theorem [Petrov '12]:** The fluctuations $\tilde{h}_N := h_N - \mathbb{E}[h_N]$ converge to a GFF in the complex structure defined by $\zeta(x, y)$:

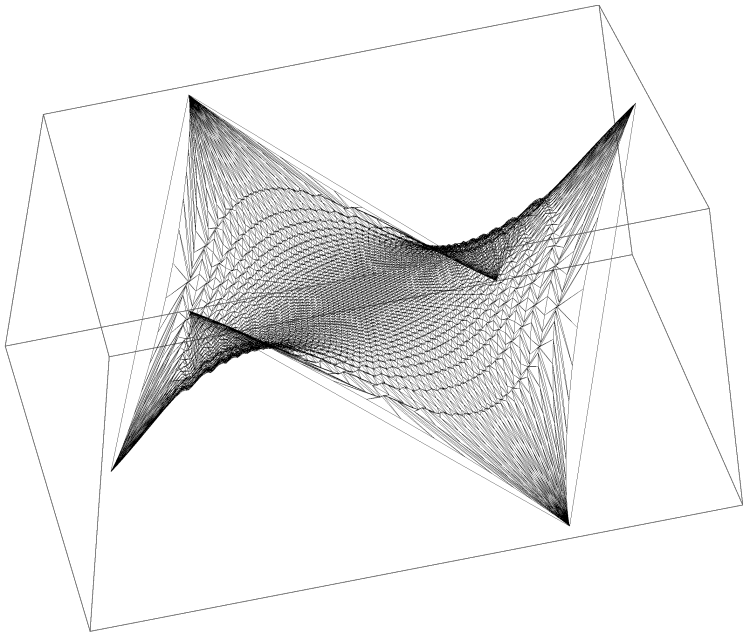
$$\tilde{h}_N \to F_{\mathbb{H}} \circ \zeta$$

where $F_{\mathbb{H}}$ is a Dirichlet GFF on $\mathbb{H}$.



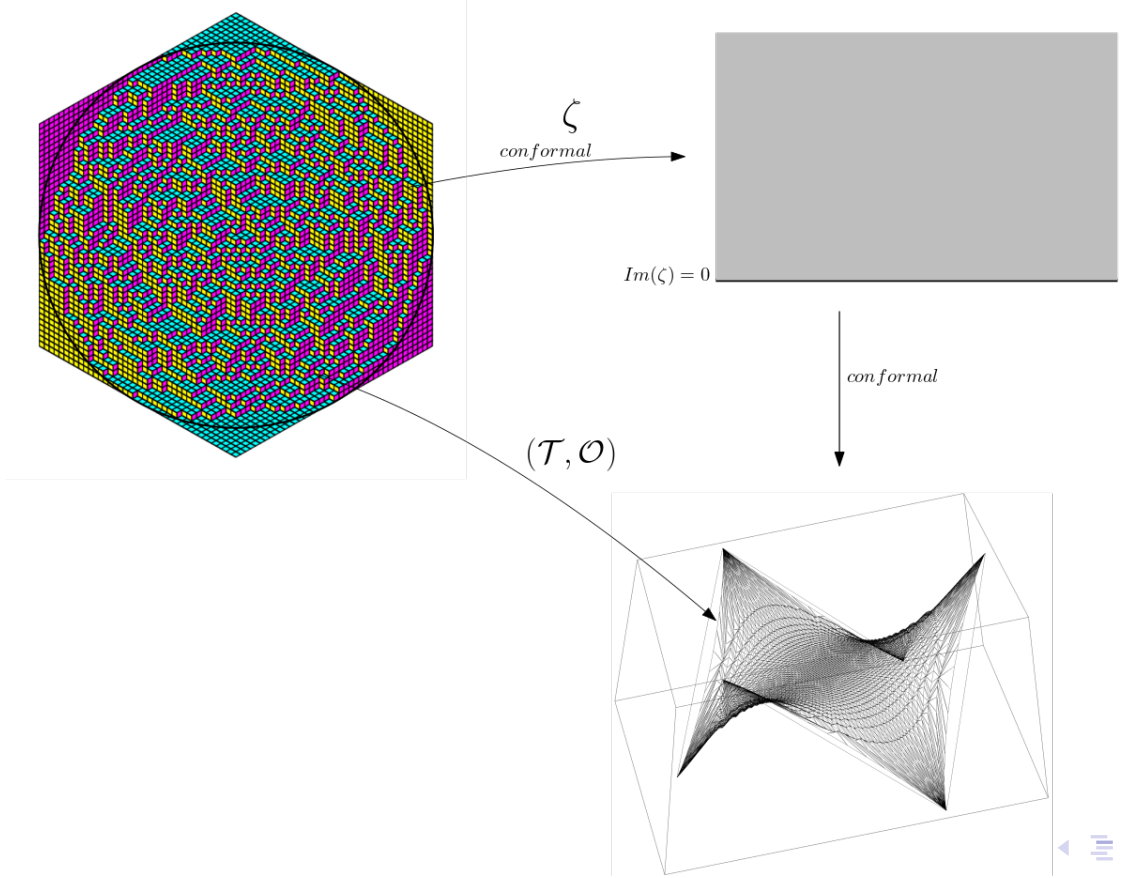
Asymptotic fluctuations via t-embeddings

1. We also prove the **two regularity conditions** required to apply [CLR '21, Theorem 1.4].
2. Our results + [CLR '21, Theorem 1.4] imply (a new proof of): height function fluctuations converge to Gaussian Free Field in the conformal structure of the surface. Is it consistent with [Pet '12]?



Asymptotic fluctuations via t-embeddings

1. Our results + [CLR '21, Theorem 1.4] imply (a new proof of): height function fluctuations converge to Gaussian Free Field in the conformal structure of the surface. Is it consistent with [Petrov '12]?
2. Our formula for $(\mathcal{T}_A(A\chi, A\eta), \mathcal{O}_A(A\chi, A\eta))$ is a composition:



The two regularity conditions

1. In our previous work [BNR '23], we showed that to get the nondegneracy assumptions for the result of [CLR '21], it suffices to prove that in compact subsets of the liquid region
   - dual edges e^* satisfy $\frac{1}{CA} \leq |d\mathcal{T}(e^*)| \leq \frac{C}{A}$, and
   - angles are bounded away from 0 and π.
2. We argue the above by deriving and analyzing an asymptotic expansion for large A

$$F(x,n) \approx \frac{1}{\sqrt{A}} \left(e^{AS(\zeta;\chi,\eta)} f(\zeta) \sqrt{\frac{\partial \zeta}{\partial \chi}} + e^{A\overline{S(\zeta;\chi,\eta)}} f(\overline{\zeta}) \sqrt{\overline{\frac{\partial \zeta}{\partial \chi}}} \right)$$

$$G(x,n) \approx \frac{1}{\sqrt{A}} \left(e^{-AS(\zeta;\chi,\eta)} g(\zeta) \sqrt{\frac{\partial \zeta}{\partial \chi}} + e^{-A\overline{S(\zeta;\chi,\eta)}} g(\overline{\zeta}) \sqrt{\overline{\frac{\partial \zeta}{\partial \chi}}} \right).$$

Pre-limit to limit correspondence

1. Recall, along a dual edge e^*, $d\mathcal{T}_A(e^*)$ and $d\mathcal{O}_A(e^*)$ can be computed via the *Coulomb gauges* F and G: What we proved is

$$\lim \sum_{(bw)^* \in \gamma^*} G(w)K(w,b)F(b) = z(\chi,\eta) = \frac{1}{2\pi \mathrm{i}} \int_{\overline{\zeta}}^{\zeta} f(z)g(z)dz$$

$$\lim \sum_{(bw)^* \in \gamma^*} \overline{G(w)}K(w,b)F(b) = \vartheta(\chi,\eta) = \frac{1}{2\pi \mathrm{i}} \int_{\overline{\zeta}}^{\zeta} f(z)\overline{g(\overline{z})}dz.$$

2. We derived an asymptotic expansion for large A

$$F(x,n) \approx \frac{1}{\sqrt{A}} \left(e^{AS(\zeta;\chi,\eta)} f(\zeta) \sqrt{\frac{\partial \zeta}{\partial \chi}} + e^{A\overline{S(\zeta;\chi,\eta)}} f(\overline{\zeta}) \sqrt{\overline{\frac{\partial \zeta}{\partial \chi}}} \right)$$

$$G(x,n) \approx \frac{1}{\sqrt{A}} \left(e^{-AS(\zeta;\chi,\eta)} g(\zeta) \sqrt{\frac{\partial \zeta}{\partial \chi}} + e^{-A\overline{S(\zeta;\chi,\eta)}} g(\overline{\zeta}) \sqrt{\overline{\frac{\partial \zeta}{\partial \chi}}} \right).$$

We expect an asymptotic of this form to hold more generally.

Open questions

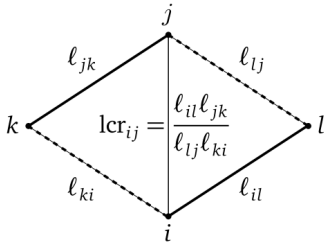
1. Existence and uniqueness? (See [CLR '21] for details and precise conjectures.)
2. What is the mechanism for the appearance of a maximal surface?
3. Can we prove existence via a variational principle? C.f. works of Thurston, Bobenko, Rivin, Colin de Verdiere.

More on a (conjectural) variational principle

Discrete conformal maps and ideal hyperbolic polyhedra

ALEXANDER I. BOBENKO
ULRICH PINKALL
BORIS A. SPRINGBORN

We establish a connection between two previously unrelated topics: a particular discrete version of conformal geometry for triangulated surfaces, and the geometry of



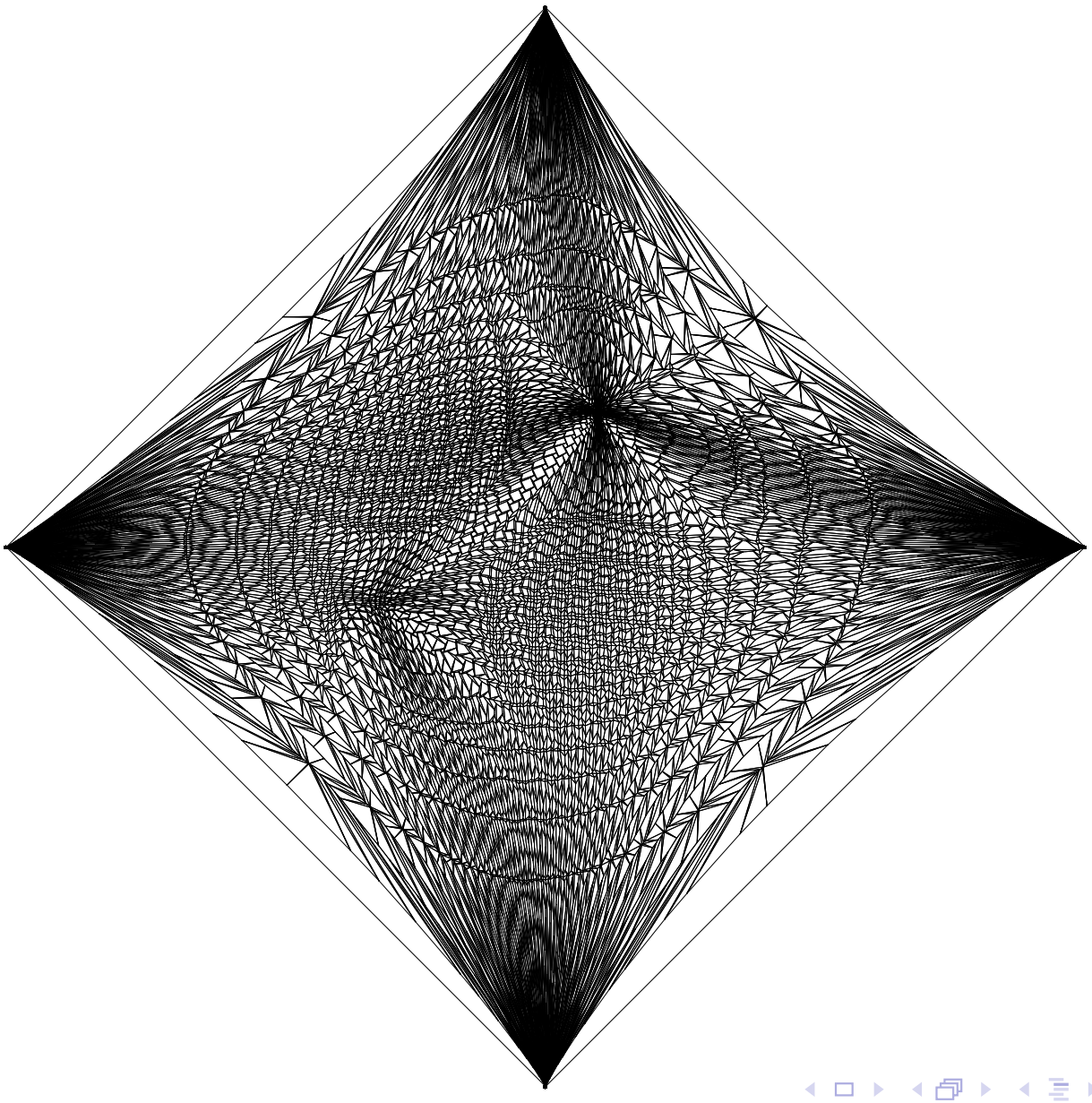
3.1.2 Problem (planar triangulation with prescribed boundary angles) ***Given***

- A euclidean triangulation (T, ℓ) that is topologically a disc and
- a desired angle sum Θ_i for each boundary vertex i,

find a discretely conformally equivalent planar triangulation $(\mathsf{T}, \tilde{\ell})$ with the given angle sums at the boundary. (The triangulated planar region may overlap with itself.)

$$E_{\mathsf{T},\Theta,\lambda}(u) = \sum_{ijk\in T} \Big(\tilde{\alpha}^k_{ij}\tilde{\lambda}_{ij} + \tilde{\alpha}^i_{jk}\tilde{\lambda}_{jk} + \tilde{\alpha}^j_{ki}\tilde{\lambda}_{ki} + 2\Pi(\tilde{\alpha}^k_{ij}) + 2\Pi(\tilde{\alpha}^i_{jk}) + 2\Pi(\tilde{\alpha}^j_{ki}) - \tfrac{\pi}{2}(\tilde{\lambda}_{ij} + \tilde{\lambda}_{jk} + \tilde{\lambda}_{ki}) \Big) + \sum_{i\in V} \Theta_i u_i \,.$$

4.1.3 Proposition (First variational principle) *Let $\mathcal{C}$ be a discrete conformal class with representative metric $\ell = e^{\lambda/2}$, and let $\tilde{\ell} = e^{\tilde{\lambda}/2}$ where $\tilde{\lambda}$ is the function of u defined by equations (2–3). Then*

- *$\tilde{\ell}$ solves Problem 3.1.1 if and only if u is a critical point of $E_{\mathsf{T},\Theta,\lambda}$,*
- *$\tilde{\ell}$ solves Problem 3.1.3 if and only if u is a critical point of $E_{\mathsf{T},\Theta,\lambda}$ with fixed u_i for $i \in V_0$. (In this case, the values of Θ for $i \in V_0$ are irrelevant.)*

End!